

# SMART AMRITA HACKATHON 2026 | POSTER PRESENTATION

## Guidelines for Students

The Poster Presentation is the research-communication component of Smart Amrita Hackathon (SAH) 2026. It invites students and research scholars to present **their own project, experimental or research outcome** on a single A2 poster before an expert jury of faculty, researchers and industry professionals. Unlike the Hackathon and the Project Expo, it requires neither a new build nor a live demonstration — it recognises technical work that can be stated, defended and discussed.

### 1. Objectives

- Provide a platform for students and research scholars to present project, laboratory and research outcomes.
- Develop the ability to communicate technical work concisely and defend it before an expert panel.
- Recognise original student contribution, including work not suited to live demonstration.
- Identify work with potential for publication, patenting or further research support.

### 2. Eligibility — Who Can Apply?

| Criterion              | Requirement                                                                                                  |
|------------------------|--------------------------------------------------------------------------------------------------------------|
| Who may apply          | Students of any B.Tech, M.Tech or PhD programme, in any year of study.                                       |
| Mode of entry          | <b>Individual.</b> Each poster carries a single author; there are no teams in this category.                 |
| Faculty mentor / guide | Mandatory. A faculty mentor or research guide must endorse the entry, certifying authorship and originality. |
| Entries per student    | <b>One poster per student.</b>                                                                               |
| Stage of work          | Completed, or sufficiently advanced to present a defensible result.                                          |
| Registration           | Free. No registration fee is charged for any SAH 2026 category.                                              |

### 3. Poster Scope

**Qualifying work.** An entry qualifies if it presents at least one of the following, authored by the student: *a mini, capstone or design project; laboratory or experimental work extended beyond the prescribed exercise; work carried out under a funded or institutional research project; a research paper published, accepted or under review; an M.Tech or PhD research outcome; internship or industry project work with written permission from the host organisation; or an independent design, analysis or replication study with original results.*

**Tracks.** Every entry is placed in one of ten thematic tracks. Tracks are thematic, not departmental — a student of any programme may enter any track, and placement follows the subject of the work rather than the department of the author.

| Code      | Track                                                | Typical scope                                                                                                                                                                            |
|-----------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>T1</b> | Smart Manufacturing and Industry 5.0                 | Machining and process optimisation, additive manufacturing, metrology and inspection, digital twins, condition monitoring, materials characterisation, industrial IoT on the shop floor. |
| <b>T2</b> | Robotics, Drones and Autonomous Systems              | Manipulators, mobile robots, UAV design and control, SLAM and navigation, perception for autonomy, human–robot interaction, actuation and mechanism design.                              |
| <b>T3</b> | Artificial Intelligence and Applied Machine Learning | Model architectures, computer vision, natural language processing, generative and foundation models, edge and efficient AI, explainability and responsible AI.                           |
| <b>T4</b> | Data Science, Analytics and Decision Intelligence    | Statistical modelling, forecasting, optimisation and operations research, data engineering, visual analytics, decision-support systems.                                                  |

| Code       | Track                                                   | Typical scope                                                                                                                                                              |
|------------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>T5</b>  | Cyber Security, Privacy and Trusted Computing           | Threat detection, cryptography and applied security, secure software and hardware, digital forensics, privacy-preserving computation, blockchain and trust infrastructure. |
| <b>T6</b>  | Next-Generation Communication, IoT and Embedded Systems | 5G/6G and wireless systems, antennas and RF, signal processing, VLSI and embedded design, sensor networks, edge computing infrastructure.                                  |
| <b>T7</b>  | Energy, Sustainability and Climate Action               | Renewable energy systems, energy storage and efficiency, thermal and fluid systems, emissions and waste, circular-economy engineering, environmental monitoring.           |
| <b>T8</b>  | Healthcare, Biomedical and Assistive Technology         | Medical devices and instrumentation, biosignal and medical image analysis, rehabilitation and assistive engineering, health informatics, biomechanics.                     |
| <b>T9</b>  | Smart Infrastructure, Mobility and Transportation       | Intelligent transport systems, electric and connected vehicles, structural and infrastructure health, smart-city systems, logistics and supply chain.                      |
| <b>T10</b> | Social Innovation, Inclusive and Educational Technology | Rural and community technology, accessibility, agri-tech, educational technology, low-cost engineering for underserved users, human-centred design.                        |

**Track allotment.** Authors indicate a preferred track at registration. The Organising Committee confirms or reassigns the track after screening, and its decision on placement is final. Reassignment is common and carries no adverse implication — it means only that the work sits better with a different jury.

**Essential condition.** The poster must carry a contribution that belongs to the author. Posters assembled from downloaded material, review-only or survey-only posters with no analysis, experiment, design or data of the author's own, product brochures, and work previously awarded at any Chennai Campus event will **not** be accepted.

#### 4. Selection Process and Key Dates

The Poster Presentation will be held on **Thursday, 10 September 2026**. All deadlines close at 5.00 p.m. on the date indicated. No extensions will be granted.

| #        | Stage                                 | Description                                                                                                                | Date                         |
|----------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>1</b> | Call for Posters & Registration Opens | Event coordinators invite students and research scholars to nominate their work. Online registration opens.                | <b>Mon, 24 Aug 2026</b>      |
| <b>2</b> | Registration Closes                   | Last date for entries, with poster title, author and mentor details. Late entries will not be considered.                  | <b>Mon, 31 Aug 2026</b>      |
| <b>3</b> | Poster Submission                     | Shortlisted authors shall submit the completed poster as a PDF, prepared strictly on the stipulated A2 template.           | <b>Sat, 5 Sep 2026</b>       |
| <b>4</b> | Screening & Shortlisting              | Screening committees review all entries for originality, technical merit, completeness and presentation readiness.         | <b>Sun–Mon, 6–7 Sep 2026</b> |
| <b>5</b> | Announcement of Shortlist             | Selected authors are notified and Poster IDs are allotted.                                                                 | <b>Mon, 7 Sep 2026</b>       |
| <b>6</b> | Poster Presentation & Jury Evaluation | Posters are displayed and presented before visitors and the expert jury, and evaluated using the rubric in Section 6.      | <b>Thu, 10 Sep 2026</b>      |
| <b>7</b> | Awards & Opportunity Mapping          | Winners announced. Work with strong potential is referred for publication, IP/patent support or further research guidance. | <b>Thu, 10 Sep 2026</b>      |

## 5. Poster Specification

**Size and layout.** A2 portrait (420 × 594 mm), two-column, only. There is no alternative size or layout. Landscape orientation and any other sheet size will not be mounted. Posters must be prepared on the official SAH 2026 A2 template issued with these guidelines; two completed worked samples — one hardware, one software — are issued alongside it for reference.

**Mandatory sections.** The template carries eight sections, all of which are mandatory: (1) *Problem Statement* · (2) *Objectives* · (3) *Methodology, as a flow or block diagram* · (4) *Experimental Setup, as a schematic of the rig, circuit, CAD model or system* · (5) *Results & Discussion, with at least one chart or quantitative table* · (6) *Novelty & Own Contribution* · (7) *Conclusion & Impact* · (8) *References, IEEE style, maximum six*. Sections may be resized; none may be deleted.

| Requirement            | Specification                                                                                                                                                                                                                                |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fonts                  | Cambria for headings, Calibri for body text. Both are used throughout the template. Do not substitute other fonts — a font absent from the print shop's machine will be replaced silently and the layout will break after printing.          |
| Type Scale             | Four sizes only. Title 36 pt Cambria bold · Section headings 24 pt Cambria bold · Author name, track badge 20 pt Calibri · All body text, bullets, captions, references and chart labels 16 pt Calibri. Do not introduce intermediate sizes. |
| Text and visuals       | Body text not exceeding 600 words in total across the whole poster. Figures, charts and diagrams to occupy at least half the sheet.                                                                                                          |
| Originality of figures | At least two figures must be authored by the student. Any figure reproduced from another source must be cited directly beneath it.                                                                                                           |
| Images                 | Minimum 300 dpi at printed size.                                                                                                                                                                                                             |
| Printing               | A2 portrait on 170 gsm matte art paper. Lamination is discouraged as it causes glare. Authors arrange their own printing; clips are provided at the venue.                                                                                   |
| File naming            | SAH2026_POSTER_[TrackCode]_[PosterID]_[Surname].pdf — single flattened PDF, fonts embedded. The editable source file is to be retained and produced on request.                                                                              |

## 6. Evaluation Rubric

Each poster is evaluated out of 50 marks during the display and technical interaction. The author presents for five minutes, followed by three minutes of jury questioning. Forty of the fifty marks are for technical substance; ten are for how well the work is defended in person.

| Criterion                                       | Marks     | What the Jury Looks For                                                                                                                                                                                  |
|-------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Problem Definition &amp; Objectives</b>      | <b>5</b>  | Clarity and relevance of the problem; well-defined, measurable objectives; appropriate scope.                                                                                                            |
| <b>Originality &amp; Own Contribution</b>       | <b>10</b> | A real contribution by the author, clearly identified in Section 6 of the poster (Novelty & Own Contribution); honest acknowledgement and citation of what was reused; the claim withstands questioning. |
| <b>Technical Content &amp; Methodology</b>      | <b>10</b> | Correctness of concepts, equations, units and terminology; soundness of method; assumptions and limitations stated; depth of understanding one step beyond what is printed.                              |
| <b>Results, Validation &amp; Outcome</b>        | <b>10</b> | Quantitative results in readable form; validation, comparison or a baseline; limitations and failure cases acknowledged rather than hidden.                                                              |
| <b>Poster Design &amp; Visual Communication</b> | <b>5</b>  | Legibility at reading distance; logical flow; figures that carry the argument; freedom from overcrowding and uncorrected errors.                                                                         |
| <b>Presentation &amp; Response to Jury</b>      | <b>10</b> | Clarity and confidence within five minutes; command of the subject in questioning; accuracy in stating what is not yet known.                                                                            |
| <b>Total</b>                                    | <b>50</b> |                                                                                                                                                                                                          |

**Note:** The author must clearly distinguish self-developed work from externally sourced datasets, libraries, software, hardware or published methods. Fabricated results, plagiarised content and undisclosed use of generative AI to produce results or the claimed contribution will lead to disqualification. Use of AI tools for language editing, figure layout or code assistance is permitted and must be declared at registration.

## 7. Awards and Recognition

Thirteen awards are conferred — one Best Poster in each of the ten tracks, and three named cohort awards so that postgraduate, doctoral and junior undergraduate entries are also recognised on their own terms. Track awards are open to every level of study.

| #  | Award                                                                 | Track / Cohort                         |
|----|-----------------------------------------------------------------------|----------------------------------------|
| 1  | Best Poster — Smart Manufacturing and Industry 5.0                    | Track T1                               |
| 2  | Best Poster — Robotics, Drones and Autonomous Systems                 | Track T2                               |
| 3  | Best Poster — Artificial Intelligence and Applied Machine Learning    | Track T3                               |
| 4  | Best Poster — Data Science, Analytics and Decision Intelligence       | Track T4                               |
| 5  | Best Poster — Cyber Security, Privacy and Trusted Computing           | Track T5                               |
| 6  | Best Poster — Next-Generation Communication, IoT and Embedded Systems | Track T6                               |
| 7  | Best Poster — Energy, Sustainability and Climate Action               | Track T7                               |
| 8  | Best Poster — Healthcare, Biomedical and Assistive Technology         | Track T8                               |
| 9  | Best Poster — Smart Infrastructure, Mobility and Transportation       | Track T9                               |
| 10 | Best Poster — Social Innovation, Inclusive and Educational Technology | Track T10                              |
| 11 | Postgraduate Poster Research Excellence Award                         | M.Tech scholars, across all tracks     |
| 12 | Doctoral Poster Research Excellence Award                             | PhD scholars, across all tracks        |
| 13 | Emerging Researcher Poster Award                                      | First-year students, across all tracks |

**Recognition.** *Each award winner receives a medal and a Certificate of Merit.* All other participants whose posters are displayed and presented to the jury receive an e-participation certificate. Certificates of Appreciation are issued to the mentors of award-winning posters.

**Award eligibility.** A poster may receive one award only; where a poster tops both a track and a cohort, it retains the track award and the cohort award passes to the next-ranked poster in that cohort. If no deserving poster is found in a track or cohort, the award will not be conferred. The decision of the jury is final.

## 8. General Information

**Display.** Board numbers follow the Poster ID and are displayed at the venue from 08.00 hrs on the event day. Mounting is on the boards provided, using the clips supplied; adhesive tape, pins and staples are not permitted. Authors are responsible for removing their posters at the close of the event.

**Presentation.** The author must be present at the poster throughout every scheduled jury round; an unattended poster is not evaluated. Presentation may be in English or Tamil; the poster itself must be in English. A laptop or tablet may be kept at the poster to show a short demonstration video on request. Live hardware demonstrations, power connections and running equipment belong to the Project Expo and are not permitted at the poster boards.

**Handouts.** Authors are advised to carry ten A4 reductions of their A2 poster for the jury and for visitors.

**Academic integrity.** Posters must be original student work. Authors must acknowledge external resources, cite software, libraries, datasets and published methods used, declare AI-assisted work as required by institutional policy, and present no fabricated results. Misrepresentation of third-party work as student-developed will lead to disqualification.

**Cross-entry.** A student may enter the same work in only one SAH 2026 category. Identical work submitted to both the Project Expo and the Poster Presentation will be withdrawn from one.

## 9. Enquiries and Organising Committee

Students may approach the Programme Faculty Coordinator for their own programme in the first instance. Queries of a general nature may be sent to the Organising Committee by email.

| # | Organising Committee Member | Programme / Role                         |
|---|-----------------------------|------------------------------------------|
| 1 | Dr. S. Parthasarathy        | Poster Presentation — Campus SPoC        |
| 2 | Dr. Rishikumar              | MECH & RAI Programme Faculty Coordinator |
| 3 | Dr. S. Veluchamy            | ECE & CCE Programme Faculty Coordinator  |
| 4 | Dr. K. Ashwini              | CSE Programme Faculty Coordinator        |
| 5 | Dr. IR Oviya                | AI & CYS Programme Faculty Coordinator   |

| Channel                | Details                                                                                                                                                               |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Email                  | sah@ch.amrita.edu                                                                                                                                                     |
| Website — full details | [ website address to be inserted ] — templates, registration link, formats, FAQs and announcements.                                                                   |
| Templates issued       | SAH2026_Poster_Template_A2.pptx (blank, customisable) with two worked samples — SAH2026_Poster_Sample_A2_Hardware.pptx and SAH2026_Poster_Sample_A2_Software_AI.pptx. |

Issued by

**The Organising Committee, Smart Amrita Hackathon (SAH) 2026**

*Amrita School of Engineering and Computing, Chennai Campus*

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